

Vinitkumar Jani

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DATA SCIENCE PROFILE

High-performing Computing Science undergraduate (Ranked 1st in Department) with a specialization in **Machine Learning and Algorithmic Optimization**. Proficient in building end-to-end predictive pipelines using **Python, SQL, and XGBoost**. Expert in handling high-dimensional datasets (13M+ records) and optimizing data structures to improve system performance by **40%**. Proven ability to apply **first-principles thinking** to solve complex analytical problems.

TECHNICAL SKILLS

- **Machine Learning:** XGBoost, Facebook Prophet, Time-Series Forecasting, Regression Analysis, Scikit-learn.
 - **Programming:** Python (Advanced), Java, Flask (Web Framework), Jinja2, NumPy, Pandas.
 - **Mathematics & Algorithms:** Computational Complexity (Big O), Data Structures (Trees, Graphs), Statistics.
 - **Data Engineering:** SQL (Star Schema Design), ETL Pipelines, SQLAlchemy, Database Optimization.
 - **Tools:** Git/GitHub, Matplotlib, Seaborn, Power BI, Linux.
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KEY DATA SCIENCE PROJECTS

VigilAI: Multi-Class Real-Time Surveillance Ecosystem

- **Objective:** Engineered an advanced surveillance system integrating the **YOLOv26** architecture with biometric face recognition and **Explainable AI (XAI)**.
- **Machine Learning:** Fine-tuned the **YOLOv26-Nano** backbone using **Transfer Learning** and robust augmentation (Mosaic & Noise injection) on a **12,000-image dataset**.
- **Performance:** Achieved a real-time throughput of **32 FPS** on consumer CPUs and an **mAP of 0.825**.
- **Interpretability:** Integrated **Grad-CAM** and **LIME** modules to provide visual justifications (heatmaps) for model classifications, addressing "black-box" AI concerns.

London Crime Forecasting & Predictive Engine (Python, XGBoost, Prophet)

- **Objective:** Developed a predictive model to forecast metropolitan crime volumes across London boroughs.
- **Feature Engineering:** Integrated external **weather datasets** (Avg Temp) and engineered **lag variables** (y_{lag1} , y_{lag12}) to capture seasonality and cyclical trends.
- **Modeling:** Deployed **XGBoost Regressor** and **Facebook Prophet** for time-series forecasting; validated models using **TimeSeriesSplit** to prevent data leakage.

- **Optimization:** Achieved significant R^2 scores by tuning hyperparameters and addressing data skewness in 13 million records.

MediTrack: Algorithmic Optimization (Python, MySQL)

- Engineered a hospital management backend where query speeds were increased by **40%** through the implementation of **QuickSort** and **MergeSort** for patient record indexing.
- Designed a relational database with normalized schemas to ensure data integrity and low-latency retrieval.

Journey Advisor (AI Travel Optimization)

- Developed a full-stack Flask application that calculates carbon footprints and travel costs.
- Implemented smart-search logic and secure session management for personalized user data analysis.

PROFESSIONAL EXPERIENCE

Victoria Solutions | London *Data Analysis & AI Intern | Jul 2025 – Sep 2025*

- **AI Implementation:** Completed specialized modules in **AI Applications**, focusing on the deployment of automation scripts for business intelligence.
- **Statistical Modeling:** Leveraged **SQL** to structure multi-source datasets for predictive executive reporting.
- **Recognition:** Awarded **Certificate of Excellence** for high-quality technical delivery and analytical standards.

G Asian Canteen | London *Part-Time IT Manager (Systems Analyst) | Jun 2024 – Aug 2024*

- **Algorithm Design:** Digitalized manual inventory tracking, utilizing logic-based scripts to improve **data accuracy by 20%**.
- **Process Improvement:** Optimized workflow systems resulting in a **25% gain in operational efficiency**.

Pho Vietnamese Street Food | London *Server / Line Chef | Jan 2024 – Jan 2026*

- **Real-time Processing:** Managed high-frequency transaction data and optimized kitchen workflows to maintain system consistency during peak demand.

EDUCATION

BSc Computing Science | Coventry University, London

- **Academic Ranking:** **1st in the Department of Computing**.
- **Core Modules:** Algorithms & Programming (Advanced Python), Data Science & Databases (SQL Optimization), Computational Complexity, AI Applications.

CERTIFICATIONS

- **Data & Business Analysis Professional** – Victoria Solutions (2025).

- **Learning SQL Programming** – LinkedIn & NASBA (2026).
- **AI Internship Certificate** – Victoria Solutions (2025).